

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2188

Westboro 100 kV
March 27, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2188. The Interconnection Customer proposes a 215.3 MW wind facility interconnecting at the Westboro 100 kV substation in the Indiana area. The request was received on September 25, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 597 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,494,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: BLUE RAVINE-SCHOOL – THOMAS HILL STATION SUBSTATION 230kV
Contingency: MONTPELIER ELECTRIC GENERATING STA – MUSKOGEE 500kV
Loading: 107.3% (Rating: 815 MVA)
DFAX: 6.2%
Pre-project loading: 105.7%
2. Facility: SHIPROCK 2 – MAKAHLOY 100kV
Contingency: WABASH 500-230 kV Transformer Bank
Loading: 102.8% (Rating: 120 MVA)
DFAX: 5.6%
Pre-project loading: 92.7%
3. Facility: WRANGELL – DEVERS 100kV
Contingency: WABASH 500 kV/230 kV XFMR
Loading: 102.8% (Rating: 113 MVA)
DFAX: 5.6%
Pre-project loading: 92.1%
4. Facility: HANDSOME LAKE ENERGY LLC – TAMUNING 100kV
Contingency: WABASH 500/230kV Auto
Loading: 100.9% (Rating: 124 MVA)
DFAX: 5.6%
Pre-project loading: 91.1%
5. Facility: DEVERS – WALSENBURG SUBSTATION 100kV
Contingency: WABASH 500-230 kV Transformer
Loading: 100.8% (Rating: 98 MVA)
DFAX: 5.6%
Pre-project loading: 88.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BLUE RAVNE-SCHOOL – THOMAS HILL STA 230kV	Reconductor line	\$14,344,000	2% (\$251,000)
MAKAHOY SUBSTATION – SHIPROCK 2 100kV L	Reconductor line	\$1,889,000	49% (\$930,000)
WRANGELL – DEVERS 100kV	Reconductor line	\$3,505,000	49% (\$1,726,000)
HANDSOME LAKE ENERGY LLC – TAMUNING 100kV	Reconductor line	\$1,601,000	49% (\$789,000)
DEVERS – WALSENBURG SUBSTATION 100kV	Reconductor line	\$1,621,000	49% (\$798,000)
	Total allocated to project		\$4,494,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.